Hmwk 10

Sunday, April 26, 2020 11:16 PM

Homework # 10

1. (50 points) The activation energy of an isomerization reaction which obeys 1st-order kinetics is known to be 100 kJ/mol and the pre-exponential of reaction is $1x10^5$ cm/min. When the isomerization reaction is run under strong pore diffusion limitations at 400 K, 500 g of catalyst are required to achieve 50% conversion.

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a) Write the mole balance for the packed-bed reactor.
b) Calculate how much catalyst is required to get 50% conversion if the temperature is raised to 425 K

You should assume that there are no external mass transfer limitations, but that strong pore diffusion limitations are present at 425 K. The catalyst density (r_c) is 2.5 g/cm^3 and the bed void fraction (e) is 0.4.

$$\rightarrow \frac{dF_a}{dV} = -\eta K'' S_a \rho_c (1-\varepsilon) C_A$$

$$F_a = C_A \nu$$

$$\rightarrow \frac{dC_A}{dV} = -\frac{\eta K'' A \cdot S_a \rho_c (1-\varepsilon) C_A}{\nu} = -AV$$

$$\rightarrow \ln\left(\frac{C_A}{C_{A0}}\right) = \frac{-\eta K'' S_a \rho_c (1-\varepsilon) V}{\nu} = -AV$$

$$\rightarrow \ln\left(\frac{C_A}{C_{A0}}\right) = -A W (1-\varepsilon)\rho_c \rightarrow \ln\left(\frac{50{,}000}{1\times10^5}\right) = -A(500)(2.5)(1-.40)$$

$$A = 0.000924 \qquad C_A = C_{A0}(1-x) = 1\times10^5(1-.5) = 50{,}000$$

→ For 50% at 425K, we need to find a new A
assume the only thing changing is the rate of reaction and η

→ calculate new K'' @ T = 425K

$$A = \frac{A_2}{A_1} = \frac{k_2'' \eta_2}{k_1 \eta_1} = \frac{k_2''(K_1''^{1/2})}{K_1''(K_2''^{1/2})}$$

$$K_2 = Ae^{-Ea/RT_2} \qquad K_1 = Ae^{-Ea/RT_1}$$

$$\frac{K_1}{K_2} = e^{-Ea/RT_1 + Ea/RT_2}$$

$$= e^{-100\times10^3/8.314(400) + 100\times10^3/8.314(425)}$$

$$= 0.17$$

$$\frac{W_2}{W_1} = \frac{K_1}{K_2} \Rightarrow W_2 = 0.17(500\,g) = 425\,g$$

2. (50 points) The irreversible gas-phase reaction

A □□□□B

is carried out over a packed bed of solid catalyst particles. The reaction is first order in the concentration of A on the catalyst surface:

$$-r_{As}' = k'C_{As}$$

The feed consists of 50% (mole) A and 50% inerts and enters the bed at a temperature of 300 K. The entering volumetric flow rate is 10 dm^3/s (i.e. 10,000 cm^3/s). The relationship between the Sherwood number and the Reynolds number is:

$$Sh = 100\ Re^{1/2}$$

As a first approximation, one may neglect pressure drop. The entering concentration of A is 1.0 M. Calculate the catalyst weight necessary to achieve 60% conversion of A for isothermal operation.

Additional Information:
Kinematic viscosity: □ = 0.02 cm^2/s
Particle diameter: d_p = 0.15 cm
Superficial velocity: U = 9 cm/s
Catalyst surface area/mass of catalyst bed: a = 60 cm^2/g
Diffusivity of A: D_e = 10^{-2} cm^2/s
k' (300 K) = 0.01 $cm^3/s/g$

$$A \rightleftharpoons B$$

$$-r_A' = k' C_{AS} \qquad Re = \frac{d_p U}{\nu}$$

$$Sh = 100 \cdot Re^{1/2} \qquad x = .60$$

$$\text{if } Re = 9\ \frac{cm}{s} \times \frac{0.15\,cm}{0.02\,cm^2/s} = 67.5$$

$$Sh = 100\,(67.5)^{1/2} = 821.50$$

$$Sh = \frac{k_c \cdot d_p}{D_{AB}} \qquad k_c = \frac{(10^{-2})(821.5)}{.15}$$

$$k_c = 54.76\ \frac{cm}{s}$$

$$K_{eff} = \frac{k_c \cdot k''}{(k_c + k'')} \Rightarrow \qquad k'' = \frac{k'}{a} = \frac{.01\,cm^3/(s\cdot g)}{60\,\frac{cm^2}{g_{cat}}} = 1.67\times10^{-4}\ \frac{cm}{s}$$

$$K_{eff} = \frac{(54.76\frac{cm}{s})(1.67\times10^{-4})}{54.76 + 1.67\times10^{-4}} = 1.67\times10^{-4}$$

$$\frac{dF_A}{dW} = K_{eff} \cdot C_A^{a} =$$

$$\ln\left(\frac{C_A}{C_{A0}}\right) = -\frac{W}{v_0}(K_{eff})\cdot a \longrightarrow \qquad x = 0.6 \quad 1 - \frac{C_A}{C_{A0}} = 0.6$$

$$\frac{\ln(0.4) \times 10{,}000\,\frac{cm^3}{s}}{(1.67\times10^{-4}\,\frac{cm}{s})(60\,\frac{cm^2}{g})} = W = 2962.8\ kg$$